Computer Programming (CSN11101)
ASSIGNMENT-7

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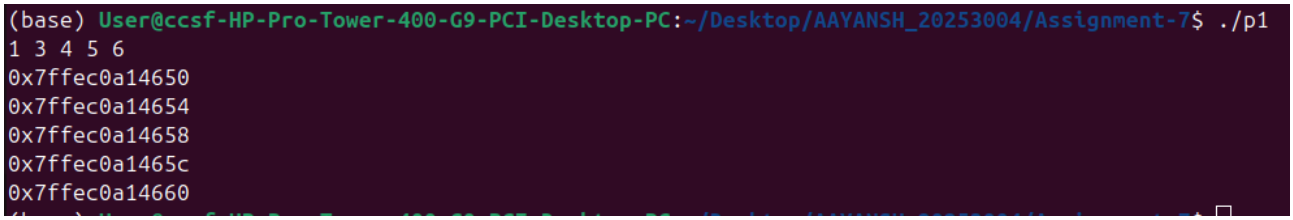
Q1. Write a C program to get array of 5 integers and print the address and value of each index one by one using pointer notations.

A1. CODE:

```
#include<stdio.h>

int main(){
  int arr[5];
  for(int i=0;i<5;i++){scanf("%d",&arr[i]);}
  for(int i=0;i<5;i++){printf("%p\n",(arr+i));}
  return 0;
}
```

OUTPUT:



Q2. Write a C program to swap two integers using call by the reference method.
A2. CODE:

```
#include <stdio.h>

void swap(int *x, int *y){
    int temp=*x;
    *x=*y;
    *y=temp;
}

int main(){
    int a,b;
    printf("Enter two integers: ");
    scanf("%d %d",&a,&b);
    printf("Before:\na=%d,b=%d\n",a,b);
    swap(&a,&b);
    printf("After swapping:\na=%d,b=%d\n",a,b);
    return 0;
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment-7$ ./p2
Enter two integers: 12 14
Before:
a=12,b=14
After swapping:
a=14,b=12
```

Q3. Write a C program to create a function named 'fun' and print its base address.
A3. CODE:

```
#include <stdio.h>

void fun() {
    printf("Inside function 'fun'\n");
}
int main() {
    printf("%p\n",fun);
    return 0;
}
```

OUTPUT:

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment-7$ ./p3
0x5797f4bed169
```

Q4.Write a C program to multiply two matrices.
A4. CODE :

```
#include <stdio.h>

int main() {
    int r1,c1,r2,c2;
    printf("Enter Dimension of first matrix: ");
    scanf("%d %d",&r1,&c1);
    printf("Enter Dimension of second matrix: ");
    scanf("%d %d",&r2,&c2);
    int a[r1][c1],b[r2][c2],r[r1][c2];
    if(c1!=r2){
        printf("Invalid input");
        return 0;
    }
    printf("Enter First matrix:\n");
    for(int i=0;i<r1;i++)
        for(int j=0;j<c1;j++) scanf("%d",&a[i][j]);
    printf("Enter Second matrix:\n");
    for(int i=0;i<r2;i++)
        for(int j=0;j<c2;j++) scanf("%d",&b[i][j]);

    for (int i=0;i<r1;i++)
        for (int j=0;j<c2;j++) r[i][j]=0;

    for (int i=0;i<r1;i++)
        for (int j=0;j<c2;j++)
            for (int k=0;k<c1;k++) r[i][j]+=a[i][k]*b[k][j];

    printf("Final Matrix:\n");
    for (int i=0;i<r1;i++){
        for(int j=0;j<c2;j++) printf("%d ",r[i][j]);
        printf("\n");
    }
    return 0;
}
```

OUTPUT :

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment-7$ ./p4
Enter rows and columns of first matrix: 3 3
Enter rows and columns of second matrix: 3 3
Enter elements of first matrix:
1 2 3
4 5 6
7 8 9
Enter elements of second matrix:
9 8 7
6 5 4
3 2 1
Resultant Matrix:
30 24 18
84 69 54
138 114 90
```

Q5. Write a C program to rotate the elements of a matrix by 90 degrees anticlockwise
A5. CODE :

```
#include <stdio.h>

int main() {
    int n;
    printf("Enter value of n: ");
    scanf("%d",&n);
    int mat[10][10];
    printf("Enter matrix:\n");
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++) scanf("%d",&mat[i][j]);
    }
    for(int i=0;i<n;i++){
        for(int j=i;j<n;j++){
            int temp=mat[i][j];
            mat[i][j]=mat[j][i];
            mat[j][i]=temp;
        }
    }
    for (int j=0;j<n;j++){
        for (int i=0;i<n/2;i++){
            int k=n-i-1;
            int temp=mat[i][j];
            mat[i][j]=mat[k][j];
            mat[k][j]=temp;
        }
    }
    printf("Answer:\n");
    for (int i=0;i<n;i++){
        for (int j=0;j<n;j++)
            printf("%d ",mat[i][j]);
        printf("\n");
    }
    return 0;
}
```

OUTPUT :

```
(base) User@ccsf-HP-Pro-Tower-400-G9-PCI-Desktop-PC:~/Desktop/AAYANSH_20253004/Assignment-7$ ./p5
Enter value of n: 4
Enter matrix:
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
Answer:
4 8 12 16
3 7 11 15
2 6 10 14
1 5 9 13
```